



SIGN-OFF REQUIRED

This assessment must be digitally signed off by an Authorised User.

General Details

Assessor	Samuel Prieto Ayllon
Assessment Date	04/07/2025
Assigned Reviewer	Samuel Prieto Ayllon
Next Review Date	04/07/2026

Assigned user to Sign Off	Oraib AlKheetanRojas
Signed Off By	This assessment has not yet been signed off
Signed Off On	--
Sign Off Comments	

Operation Assessed	Dexterous Hand RH56 Series Operation
--------------------	--------------------------------------

Associated with specific area

-  Basement 2
-  B2 029 (Robotics Lab)
-  CTP

Description of work area and/or activity assessed

This assessment covers routine bench-top and robot-mounted use of the Inspire-Robots Dexterous Hand RH56, which grips objects with six linear servo actuators and integrated force sensors while powered from a 24 V DC supply. Tasks include installation, electrical connection, calibration, and object manipulation under standard laboratory conditions.

Reason for Risk Assessment

To document and control any health, safety, and equipment-damage risks introduced by actively using the RH56 dexterous hand in research.

Overall Current Risk

Moderate Risk



Average Number of Persons Affected

2	 Employees
2	 Students
2	 Visitors

Current Rating	Hazard Information	Task Status
Moderate Risk 	Electricity The hand operates at 24 V DC (quiescent 0.12 A, up to 2 A under maximum load); damaged insulation or incorrect wiring could expose conductors and cause mild electric shock or short circuit.	0 0 0
	Measures Currently in place to prevent risk of injury	
	• Supply limited to Safety Extra-Low Voltage (SELV). • Power applied only after mechanical mounting. • Visual inspection of GX12 cable and connector before each use. • Wiring changes performed solely by trained personnel.	

Potential Rating	Additional Controls Required
Moderate Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Moderate Risk 	Moving Parts The six motor-driven fingers can move quickly when commanded, producing up to 15 N on the thumb and 10 N on other fingers and potentially striking or crushing skin if body parts enter the workspace.	0 0 0
	Measures Currently in place to prevent risk of injury	
	• Software limits and reduced speed during setup. • Host-controller emergency-stop function. • Keep hands clear during actuation, operate at low speed for teaching. • Secure mounting before energising.	

Potential Rating	Additional Controls Required
Moderate Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Moderate Risk 	Pinch Points Small gaps between finger segments close completely during gripping, creating pinch points that can trap fingers, cables, or loose clothing, especially in calibration routines.	0 0 0
	Measures Currently in place to prevent risk of injury	
	• Operators wear snug-fitting gloves and avoid loose clothing. • Force-control threshold configured to stop motion on contact. • Clear verbal warning before each motion.	

Potential Rating	Additional Controls Required
Moderate Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

